CSP Assignment Problem: University Course Scheduling

Problem Statement:

A small university wants to schedule a set of 5 courses for a semester. Each course needs to be assigned:

- A time slot (from a limited list)
- A room (from a list of available rooms)
- A professor (from a list of available professors)

The goal is to assign a time, room, and professor to each course such that all constraints are satisfied.

Given:

- Courses: CS101, CS102, CS103, CS104, CS105
- Time slots: Mon 9am, Mon 11am, Tue 9am, Tue 11am, Wed 9am
- Rooms: RoomA, RoomB, RoomC
- Professors: ProfA, ProfB, ProfC

Constraints:

1. No room conflict: No two courses can be scheduled in the same room at the same time. 2. No professor conflict: A professor cannot teach two courses at the same time. 3. Availability constraint:
   - o ProfA is not available on Mon 9am.
   - o ProfB is not available on Tue 9am.
   - o ProfC is not available on Wed 9am.
4. Course-specific room requirement:
   - o CS105 must be in RoomC (it needs lab equipment).
5. Course-specific professor assignment:
   - o ProfA must teach CS101.
   - o ProfB must teach CS103.

Assignment Tasks:

1. Model the problem as a CSP with variables, domains, and constraints. 2. Implement a solution in Python (you can use backtracking or libraries like python constraint or OR-Tools).
3. Print a valid schedule showing Course, Time, Room, and

Professor. Bonus Task (Optional):

- 🎬 Add a soft constraint: Try to minimize the number of total time slots used (i.e., try to group courses into fewer time slots if possible).

Example Output Format:

Course | Time | Room | Professor
--------------------------------------- CS101 | Mon 11am | RoomA | ProfA CS102 | Tue 9am | RoomB | ProfC CS103 | Wed 9am | RoomA | ProfB CS104 | Mon 9am | RoomB | ProfC CS105 | Tue 11am | RoomC | ProfA